

MODEL EVALUATION GUIDELINES FOR SYSTEMATIC QUANTIFICATION OF ACCURACY IN WATERSHED SIMULATIONS

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ABSTRACT. Watershed models are powerful tools for simulating the effect of watershed processes and management on soil and water resources. However, no comprehensive guidance is available to facilitate model evaluation in terms of the accuracy of simulated data compared to measured flow and constituent values. Thus, the objectives of this research were to: (1) determine recommended model evaluation techniques (statistical and graphical), (2) review reported ranges of values and corresponding performance ratings for the recommended statistics, and (3) establish guidelines for model evaluation based on the review results and project-specific considerations; all of these objectives focus on simulation of streamflow and transport of sediment and nutrients. These objectives were achieved with a thorough review of relevant literature on model application and recommended model evaluation methods. Based on this analysis, we recommend that three quantitative statistics, Nash-Sutcliffe efficiency (NSE), percent bias (PBIAS), and ratio of the root mean square error to the standard deviation of measured data (RSR), in addition to the graphical techniques, be used in model evaluation. The following model evaluation performance ratings were established for each recommended statistic. In general, model simulation can be judged as satisfactory if $NSE > 0.50$ and $RSR \leq 0.70$, and if $PBIAS \pm 25\%$ for streamflow, $PBIAS \pm 55\%$ for sediment, and $PBIAS \pm 70\%$ for N and P. For PBIAS, constituent-specific performance ratings were determined based on uncertainty of measured data. Additional considerations related to model evaluation guidelines are also discussed. These considerations include: single-event simulation, quality and quantity of measured data, model calibration procedure, evaluation time step, and project scope and magnitude. A case study illustrating the application of the model evaluation guidelines is also provided.

Keywords. Accuracy, Model calibration and validation, Simulation, Watershed model.

Computer-based watershed models can save time and money because of their ability to perform long-term simulation of the effects of watershed processes and management activities on water quality, water quantity, and soil quality. These models also facilitate the simulation of various conservation program effects and aid policy design to mitigate water and soil quality degradation by determining suitable conservation programs for particular watersheds and agronomic settings. In order to use model outputs for tasks ranging from regulation to research, models should be scientifically sound, robust, and defensible (U.S. EPA, 2002).

Sensitivity analysis is the process of determining the rate of change in model output with respect to changes in model inputs (parameters). It is a necessary process to identify key parameters and parameter precision required for calibration (Ma et al., 2000). Model calibration is the process of estimating model parameters by comparing model predictions (output) for a given set of assumed conditions with observed data for the same conditions. Model validation involves running a model using input parameters measured or determined during the calibration process. According to Refsgaard (1997), model validation is the process of demonstrating that a given site-specific model is capable of making “sufficiently accurate” simulations, although “sufficiently accurate” can vary based on project goals. According to the U.S. EPA (2002), the process used to accept, reject, or qualify model results should be established and documented before beginning model evaluation. Although ASCE (1993) emphasized the need to clearly define model evaluation criteria, no commonly accepted guidance has been established, but specific statistics and performance ratings for their use have been developed and used for model evaluation (Donigian et al., 1983; Ramanarayanan et al., 1997; Gupta et al., 1999; Motovilov et al., 1999; Saleh et al., 2000; Santhi et al., 2001; Singh et al., 2004; Bracmort et al., 2006; Van Liew et al., 2007). However, these performance ratings are model and project specific. Standardized guidelines are needed to establish a common system for judging model performance and comparing various models (ASCE, 1993). Once established, these guidelines will assist modelers in preparing and reviewing quality assurance project plans for modeling (U.S. EPA, 2002) and will increase accountability and public acceptance of models to support

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scientific research and to guide policy, regulatory, and management decision-making.

A number of publications have addressed model evaluation statistics (Willmott, 1981; ASCE, 1993; Legates and McCabe, 1999), but they do not include recently developed statistics (e.g., Wang and Melesse, 2005; Parker et al., 2006). More importantly, none of the previous publications provide guidance on acceptable ranges of values for each statistic. Borah and Bera (2004) present an excellent review of values for various statistics used in hydrologic and nonpoint-source model applications, but more elaborate analysis is needed to aid modelers in determining performance ratings for these statistics.

In most watershed modeling projects, model output is compared to corresponding measured data with the assumption that all error variance is contained within the predicted values and that observed values are error free. In discussions of model evaluation statistics, Willmott (1981) and ASCE (1993) do recognize that measured data are not error free, but measurement error is not considered in their recommendations perhaps because of the relative lack of data on measurement uncertainty. However, uncertainty estimates for measured streamflow and water quality data have recently become available (Harmel et al., 2006) and should be considered when calibrating, validating, and evaluating watershed models because of differences in inherent uncertainty between measured flow, sediment, and nutrient data.

The importance of standardized guidelines for model evaluation is illustrated by the Conservation Effects Assessment Project Watershed Assessment Study (CEAP-WAS, 2005). The CEAP-WAS seeks to quantify the environmental benefits of conservation practices supported by USDA in the 2002 Farm Bill, also known as the Farm Security and Rural Investment Act. One of the CEAP-WAS goals is to formulate guidelines for calibration, validation, and application of models used in CEAP to simulate environmental effects of conservation practices. Thus, based on the need for standardized model evaluation guidelines to support watershed modeling in CEAP-WAS and other projects, the objectives for the present research were to: (1) determine recommended model evaluation techniques (statistical and graphical), (2) review reported ranges of values and corresponding performance ratings for the recommended statistics, and (3) establish guidelines for model evaluation based on the review results and project-specific considerations. In addition, a case study illustrating the application of the model evaluation guidelines was provided. This research focuses on watershed model evaluation guidelines for streamflow, sediments, and nutrients. Throughout this article, "model evaluation" refers to the applicable steps of sensitivity analysis, calibration, validation, uncertainty analysis, and application.

METHODS

MODEL EVALUATION TECHNIQUES

To determine recommended techniques for watershed model evaluation, an extensive review was conducted on published literature related to calibration, validation, and application of watershed models. Specifically, the information compiled focused on the strengths and weaknesses of each statistical and graphical technique and on recommendations for their application. The recommended model evaluation

statistics were selected based on the following factors: (1) robustness in terms of applicability to various constituents, models, and climatic conditions; (2) commonly used, accepted, and recommended in published literature; and (3) identified strengths in model evaluation. The trade-off between long-term bias and residual variance was also considered, as recommended by Boyle et al. (2000). Bias measures the average tendency of the simulated constituent values to be larger or smaller than the measured data. Residual variance is the difference between the measured and simulated values, often estimated by the residual mean square or root mean square error (RMSE). According to Boyle et al. (2000), optimizing RMSE during model calibration may give small error variance but at the expense of significant model bias. The compilation of recommended statistics was also constrained by the recommendation of Legates and McCabe (1999) to include at least one dimensionless statistic and one absolute error index statistic with additional information such as the standard deviation of measured data, and to include at least one graphical technique as well.

REPORTED VALUE RANGES AND PERFORMANCE RATINGS FOR RECOMMENDED STATISTICS

Additional literature review was conducted to determine published ranges of values and performance ratings for recommended model evaluation statistics. Reported daily and monthly values during the calibration and validation periods for streamflow, sediment, and nutrients are recorded along with the model used for evaluation in tables A-1 through A-9 in the Appendix. All the reported data were analyzed and compiled into a summary of daily and monthly value ranges for different constituents during calibration and validation (table 1). The summary values in table 1 include the sample size of the reported values (n) and the minimum, maximum, and median of the values reported for streamflow, surface runoff, sediment, organic, mineral and total nitrogen, and organic, mineral, and total phosphorus.

MODEL EVALUATION GUIDELINES

General model evaluation guidelines that consider the recommended model evaluation statistics with corresponding performance ratings and appropriate graphical analyses were then established. A calibration procedure chart for flow, sediment, and nutrients, similar to the one proposed by Santhi et al. (2001), is included to assist in application of the model evaluation guidelines to manual model calibration. It is noted, however, that these guidelines should be adjusted by the modeler based on additional considerations such as: single-event simulation, quality and quantity of measured data, model calibration procedure, evaluation time step, and project scope and magnitude. Additionally, a brief discussion of the implications of unmet performance ratings is provided.

RESULTS AND DISCUSSION

MODEL EVALUATION TECHNIQUES

Both statistical and graphical model evaluation techniques were reviewed. The quantitative statistics were divided into three major categories: standard regression, dimensionless, and error index. Standard regression statistics determine the strength of the linear relationship between simulated and measured data. Dimensionless techniques provide

a relative model evaluation assessment, and error indices quantify the deviation in the units of the data of interest (Legates and McCabe, 1999). A brief discussion of numerous model evaluation statistics (both recommended statistics and statistics not selected for recommendation) appears subsequently; however, the relevant calculations are provided only for the recommended statistics.

Several graphical techniques are also described briefly because graphical techniques provide a visual comparison of simulated and measured constituent data and a first overview of model performance (ASCE, 1993) and are essential to appropriate model evaluation (Legates and McCabe, 1999). Based on recommendations by ASCE (1993) and Legates and McCabe (1999), we recommend that both graphical techniques and quantitative statistics be used in model evaluation.

MODEL EVALUATION STATISTICS (STANDARD REGRESSION)

Slope and y-intercept: The slope and y-intercept of the best-fit regression line can indicate how well simulated data match measured data. The slope indicates the relative relationship between simulated and measured values. The y-intercept indicates the presence of a lag or lead between model predictions and measured data, or that the data sets are not perfectly aligned. A slope of 1 and y-intercept of 0 indicate that the model perfectly reproduces the magnitudes of measured data (Willmott, 1981). The slope and y-intercept are commonly examined under the assumption that measured and simulated values are linearly related, which implies that all of the error variance is contained in simulated values and that measured data are error free (Willmott, 1981). In reality, measured data are rarely, if ever, error free. Harmel et al. (2006) showed that substantial uncertainty in reported water quality data can result when individual errors from all procedural data collection categories are considered. Therefore, care needs to be taken while using regression statistics for model evaluation.

Pearson's correlation coefficient (r) and coefficient of determination (R²): Pearson's correlation coefficient (r) and coefficient of determination (R²) describe the degree of collinearity between simulated and measured data. The correlation coefficient, which ranges from -1 to 1, is an index of the degree of linear relationship between observed and simulated data. If r = 0, no linear relationship exists. If r = 1 or -1, a perfect positive or negative linear relationship exists. Similarly, R² describes the proportion of the variance in measured data explained by the model. R² ranges from 0 to 1, with higher values indicating less error variance, and typically values greater than 0.5 are considered acceptable (Santhi et al., 2001, Van Liew et al., 2003). Although r and R² have been widely used for model evaluation, these statistics are oversensitive to high extreme values (outliers) and insensitive to additive and proportional differences between model predictions and measured data (Legates and McCabe, 1999).

MODEL EVALUATION STATISTICS (DIMENSIONLESS)

Index of agreement (d): The index of agreement (d) was developed by Willmott (1981) as a standardized measure of the degree of model prediction error and varies between 0 and 1. A computed value of 1 indicates a perfect agreement between the measured and predicted values, and 0 indicates no agreement at all (Willmott, 1981). The index of agreement

represents the ratio between the mean square error and the "potential error" (Willmott, 1984). The author defined potential error as the sum of the squared absolute values of the distances from the predicted values to the mean observed value and distances from the observed values to the mean observed value. The index of agreement can detect additive and proportional differences in the observed and simulated means and variances; however, d is overly sensitive to extreme values due to the squared differences (Legates and McCabe, 1999). Legates and McCabe (1999) suggested a modified index of agreement (d1) that is less sensitive to high extreme values because errors and differences are given appropriate weighting by using the absolute value of the difference instead of using the squared differences. Although d1 has been proposed as an improved statistic, its limited use in the literature has not provided extensive information on value ranges.

Nash-Sutcliffe efficiency (NSE): The Nash-Sutcliffe efficiency (NSE) is a normalized statistic that determines the relative magnitude of the residual variance ("noise") compared to the measured data variance ("information") (Nash and Sutcliffe, 1970). NSE indicates how well the plot of observed versus simulated data fits the 1:1 line. NSE is computed as shown in equation 1:

$$NSE = 1 - \frac{\sum_{i=1}^n (Y_i^{obs} - Y_i^{sim})^2}{\sum_{i=1}^n (Y_i^{obs} - Y^{mean})^2} \quad (1)$$

where Y_i^{obs} is the *i*th observation for the constituent being evaluated, Y_i^{sim} is the *i*th simulated value for the constituent being evaluated, Y^{mean} is the mean of observed data for the constituent being evaluated, and *n* is the total number of observations.

NSE ranges between $-\infty$ and 1.0 (1 inclusive), with NSE = 1 being the optimal value. Values between 0.0 and 1.0 are generally viewed as acceptable levels of performance, whereas values ≤ 0.0 indicates that the mean observed value is a better predictor than the simulated value, which indicates unacceptable performance.

NSE was recommended for two major reasons: (1) it is recommended for use by ASCE (1993) and Legates and McCabe (1999), and (2) it is very commonly used, which provides extensive information on reported values. Sevat and Dezetter (1991) also found NSE to be the best objective function for reflecting the overall fit of a hydrograph. Legates and McCabe (1999) suggested a modified NSE that is less sensitive to high extreme values due to the squared differences, but that modified version was not selected because of its limited use and resulting relative lack of reported values.

Persistence model efficiency (PME): Persistence model efficiency (PME) is a normalized model evaluation statistic that quantifies the relative magnitude of the residual variance ("noise") to the variance of the errors obtained by the use of a simple persistence model (Gupta et al., 1999). PME ranges from 0 to 1, with PME = 1 being the optimal value. PME values should be larger than 0.0 to indicate "minimally acceptable" model performance (Gupta et al., 1999). The power of PME is derived from its comparison of model performance with a simple persistence forecast model. According to Gupta et al. (1999), PME is capable of clearly indicating poor model

performance, but it has been used only occasionally in the literature, so a range of reported values is not available.

Prediction efficiency (P_e): Prediction efficiency (P_e), as explained by Santhi et al. (2001), is the coefficient of determination (R^2) calculated by regressing the rank (descending) of observed versus simulated constituent values for a given time step. P_e determines how well the probability distributions of simulated and observed data fit each other. However, it has not been used frequently enough to provide extensive information on ranges of values. In addition, it may not account for seasonal bias.

Performance virtue statistic (PV_k): The performance virtue statistic (PV_k) is the weighted average of the Nash-Sutcliffe coefficients, deviations of volume, and error functions across all flow gauging stations within the watershed of interest (Wang and Melesse, 2005). PV_k was developed to assess if watershed models can satisfactorily predict all aspects (profile, volume, and peak) of observed flow hydrographs for watersheds with more than one gauging station (Wang and Melesse, 2005). PV_k can range from $-\infty$ to 1.0, with a PV_k value of 1.0 indicating that the model exactly simulates all three aspects of observed flow for all gauging stations within the watershed. A negative PV_k value indicates that the average of observed streamflow values is better than simulated streamflows (Wang and Melesse, 2005). PV_k was developed for use in snow-fed watersheds; therefore, it may be necessary to make adjustments to this statistic for rain-fed watersheds. PV_k was not selected for recommendation because it was developed for streamflow only. In addition, PV_k was only recently developed; thus, extensive information on value ranges is not available.

Logarithmic transformation variable (e): The logarithmic transformation variable (e) is the logarithm of the predicted/observed data ratio (E) that was developed to address the sensitivity of the watershed-scale pesticide model error index (E) to the estimated pesticide application rates (Parker et al., 2006). The value of e is centered on zero, is symmetrical in under- or overprediction, and is approximately normally distributed (Parker et al., 2006). If the simulated and measured data are in complete agreement, then $e = 0$ and $E = 1.0$. Values of $e < 0$ are indicative of underprediction; values > 0 are indicative of overprediction. Although e has great potential as an improved statistical technique for assessing model accuracy, it was not selected because of its recent development and limited testing and application.

MODEL EVALUATION STATISTICS (ERROR INDEX)

MAE, MSE, and RMSE: Several error indices are commonly used in model evaluation. These include mean absolute error (MAE), mean square error (MSE), and root mean square error (RMSE). These indices are valuable because they indicate error in the units (or squared units) of the constituent of interest, which aids in analysis of the results. RMSE, MAE, and MSE values of 0 indicate a perfect fit. Singh et al. (2004) state that RMSE and MAE values less than half the standard deviation of the measured data may be considered low and that either is appropriate for model evaluation. A standardized version of the RMSE was selected for recommendation and is described later in this section.

Percent bias (PBIAS): Percent bias (PBIAS) measures the average tendency of the simulated data to be larger or smaller than their observed counterparts (Gupta et al., 1999).

The optimal value of PBIAS is 0.0, with low-magnitude values indicating accurate model simulation. Positive values indicate model underestimation bias, and negative values indicate model overestimation bias (Gupta et al., 1999). PBIAS is calculated with equation 2:

$$PBIAS = \left[\frac{\sum_{i=1}^n (Y_i^{obs} - Y_i^{sim}) * (100)}{\sum_{i=1}^n (Y_i^{obs})} \right] \quad (2)$$

where PBIAS is the deviation of data being evaluated, expressed as a percentage.

Percent streamflow volume error (PVE; Singh et al., 2004), prediction error (PE; Fernandez et al., 2005), and percent deviation of streamflow volume (D_v) are calculated in a similar manner as PBIAS. The deviation term (D_v) is used to evaluate the accumulation of differences in streamflow volume between simulated and measured data for a particular period of analysis.

PBIAS was selected for recommendation for several reasons: (1) D_v was recommended by ASCE (1993), (2) D_v is commonly used to quantify water balance errors and its use can easily be extended to load errors, and (3) PBIAS has the ability to clearly indicate poor model performance (Gupta et al., 1999). PBIAS values for streamflow tend to vary more, among different autocalibration methods, during dry years than during wet years (Gupta et al., 1999). This fact should be considered when attempting to do a split-sample evaluation, one for calibration and one for validation.

RMSE-observations standard deviation ratio (RSR): RMSE is one of the commonly used error index statistics (Chu and Shirmohammadi, 2004; Singh et al., 2004; Vasquez-Amabile and Engel, 2005). Although it is commonly accepted that the lower the RMSE the better the model performance, only Singh et al. (2004) have published a guideline to qualify what is considered a low RMSE based on the observations standard deviation. Based on the recommendation by Singh et al. (2004), a model evaluation statistic, named the RMSE-observations standard deviation ratio (RSR), was developed. RSR standardizes RMSE using the observations standard deviation, and it combines both an error index and the additional information recommended by Legates and McCabe (1999). RSR is calculated as the ratio of the RMSE and standard deviation of measured data, as shown in equation 3:

$$RSR = \frac{RMSE}{STDEV_{obs}} = \frac{\left[\sqrt{\sum_{i=1}^n (Y_i^{obs} - Y_i^{sim})^2} \right]}{\left[\sqrt{\sum_{i=1}^n (Y_i^{obs} - Y^{mean})^2} \right]} \quad (3)$$

RSR incorporates the benefits of error index statistics and includes a scaling/normalization factor, so that the resulting statistic and reported values can apply to various constituents. RSR varies from the optimal value of 0, which indicates zero RMSE or residual variation and therefore perfect model simulation, to a large positive value. The lower RSR, the lower the RMSE, and the better the model simulation performance.

Daily root-mean square (DRMS): The daily root-mean square (DRMS), which is a specific application of the RMSE, computes the standard deviation of the model prediction error (difference between measured and simulated values). The smaller the DRMS value, the better the model performance (Gupta et al., 1999). Gupta et al. (1999) determined that DRMS increased with wetness of the year, indicating that the forecast error variance is larger for higher flows. According to Gupta et al. (1999), DRMS had limited ability to clearly indicate poor model performance.

GRAPHICAL TECHNIQUES

Graphical techniques provide a visual comparison of simulated and measured constituent data and a first overview of model performance (ASCE, 1993). According to Legates and McCabe (1999), graphical techniques are essential to appropriate model evaluation. Two commonly used graphical

techniques, hydrographs and percent exceedance probability curves, are especially valuable. Other graphical techniques, such as bar graphs and box plots, can also be used to examine seasonal variations and data distributions.

A hydrograph is a time series plot of predicted and measured flow throughout the calibration and validation periods. Hydrographs help identify model bias (ASCE, 1993) and can identify differences in timing and magnitude of peak flows and the shape of recession curves.

Percent exceedance probability curves, which often are daily flow duration curves, can illustrate how well the model reproduces the frequency of measured daily flows throughout the calibration and validation periods (Van Liew et al., 2007). General agreement between observed and simulated frequencies for the desired constituent indicates adequate simulation over the range of the conditions examined (Singh et al., 2004).

Table 1. Summary statistics from the literature review of reported NSE and PBIAS values.^[a]

Constituent	Statistic	Calibration				Validation			
		NSE		PBIAS		NSE		PBIAS	
		Daily	Monthly	Daily	Monthly	Daily	Monthly	Daily	Monthly
Streamflow	<i>n</i>	92	33	72	0	128	70	82	0
	Minimum	-0.23	0.14	-91.70	na	-1.81	-3.35	-155.60	na
	Maximum	0.95	0.91	26.50	na	0.89	0.93	47.18	na
	Median	0.89	0.79	-1.30	na	0.67	0.63	-1.90	na
Surface runoff	<i>n</i>	0	2	0	0	0	2	0	0
	Minimum	na	0.35	na	na	na	0.63	na	na
	Maximum	na	0.62	na	na	na	0.77	na	na
	Median	na	0.49	na	na	na	0.70	na	na
Sediment	<i>n</i>	2	6	0	0	2	6	0	0
	Minimum	-2.50	0.49	na	na	-3.51	-2.46	na	na
	Maximum	0.11	0.86	na	na	0.23	0.88	na	na
	Median	-1.20	0.76	na	na	-1.64	0.64	na	na
Organic nitrogen (organic N)	<i>n</i>	0	2	0	0	0	2	0	0
	Minimum	na	0.57	na	na	na	0.43	na	na
	Maximum	na	0.58	na	na	na	0.73	na	na
	Median	na	0.58	na	na	na	0.58	na	na
Mineral nitrogen (NO ₃ -N)	<i>n</i>	0	2	0	0	0	2	0	0
	Minimum	na	-0.08	na	na	na	0.64	na	na
	Maximum	na	0.59	na	na	na	0.75	na	na
	Median	na	0.26	na	na	na	0.70	na	na
Total nitrogen (organic N + NO ₃ -N)	<i>n</i>	0	0	0	0	1	6	0	0
	Minimum	na	na	na	na	0.19	0.10	na	na
	Maximum	na	na	na	na	0.19	0.85	na	na
	Median	na	na	na	na	0.19	0.76	na	na
Organic phosphorus (organic P)	<i>n</i>	0	2	0	0	0	2	0	0
	Minimum	na	0.59	na	na	na	0.39	na	na
	Maximum	na	0.70	na	na	na	0.72	na	na
	Median	na	0.65	na	na	na	0.56	na	na
Mineral phosphorus (PO ₄ -P)	<i>n</i>	0	3	0	0	0	3	0	0
	Minimum	na	0.53	na	na	na	0.51	na	na
	Maximum	na	0.78	na	na	na	0.81	na	na
	Median	na	0.59	na	na	na	0.53	na	na
Total phosphorus (organic P + PO ₄ -P)	<i>n</i>	0	1	0	0	0	1	0	0
	Minimum	na	0.51	na	na	na	0.37	na	na
	Maximum	na	0.51	na	na	na	0.37	na	na
	Median	na	0.51	na	na	na	0.37	na	na

^[a] *n* = number of reported values for the studies reviewed (sample size), NSE = Nash-Sutcliffe efficiency, PBIAS = percent bias, and na = not available (used when *n* = 0).

REPORTED RANGES OF VALUES FOR RECOMMENDED STATISTICS

A review of published literature produced ranges of daily and monthly NSE and PBIAS values for both model calibration and validation for surface runoff, streamflow, and selected constituents, including: sediment, organic nitrogen, mineral nitrogen, total nitrogen, organic phosphorus, and mineral phosphorus (tables A-1 through A-9). A few weekly values reported by Narasimhan et al. (2005) are also included to indicate ranges of values for a weekly time step. It is important to note that calibration and validation were performed for different time periods; thus, the reported values reflect this difference. A summary of reported values for each constituent appears in table 1. Median values instead of means are included because medians are less sensitive to extreme values and are better indicators for highly skewed distributions.

Most of the reviewed studies performed model calibration and validation based on streamflow (table 1). This is attributed to the relative abundance of measured long-term streamflow data compared to sediment or nutrient data. Based on the summary information for streamflow calibration and validation, daily NSE values tended to be higher than monthly values, which contradicts findings from some individual studies (e.g., Fernandez et al., 2005; Singh et al., 2005; Van Liew et al., 2007). This anomaly is potentially due to the increased sample sizes (n) for daily data. As expected, NSE and PBIAS values for streamflow were better for the calibration periods than the validation periods.

REPORTED PERFORMANCE RATINGS FOR RECOMMENDED STATISTICS

A review of published literature resulted in the performance ratings for NSE and PBIAS shown in tables 2 and 3. Because RSR was developed in this research, similar reported performance ratings are not available. Therefore, RSR ratings were based on Singh et al. (2004) recommendations that RMSE values less than half the standard deviation of measured data can be considered low. In this study, we used the recommended less than 0.5 RSR value as the most stringent ("very good") rating and suggested two less stringent ratings of 10% and 20% greater than this value for the "good" and "satisfactory" ratings, respectively.

MODEL EVALUATION GUIDELINES BASED ON PERFORMANCE RATINGS

General model evaluation guidelines, for a monthly time step, were developed based on performance ratings for the recommended statistics and on project-specific considerations. As stated previously, graphical techniques, especially hydrographs and percent exceedance probability curves, provide visual model evaluation overviews. Utilizing these important techniques should typically be the first step in model evaluation. A general visual agreement between observed and simulated constituent data indicates adequate calibration and validation over the range of the constituent being simulated (Singh et al., 2004).

Table 2. Reported performance ratings for NSE.

Model	Value	Performance Rating	Modeling Phase	Reference
HSPF	>0.80	Satisfactory	Calibration and validation	Donigian et al. (1983)
APEX	>0.40	Satisfactory	Calibration and validation (daily)	Ramanarayanan et al. (1997)
SAC-SMA	<0.70	Poor	Autocalibration	Gupta et al. (1999)
SAC-SMA	>0.80	Efficient	Autocalibration	Gupta et al. (1999)
DHM	>0.75	Good	Calibration and validation	Motovilov et al. (1999) ^[a]
DHM	0.36 to 0.75	Satisfactory	Calibration and validation	Motovilov et al. (1999) ^[a]
DHM	<0.36	Unsatisfactory	Calibration and validation	Motovilov et al. (1999) ^[a]
SWAT	>0.65	Very good	Calibration and validation	Saleh et al. (2000)
SWAT	0.54 to 0.65	Adequate	Calibration and validation	Saleh et al. (2000)
SWAT	>0.50	Satisfactory	Calibration and validation	Santhi et al. (2001); adapted by Bracmort et al. (2006)
SWAT and HSPF	>0.65	Satisfactory	Calibration and validation	Singh et al. (2004); adapted by Narasimhan et al. (2005)

^[a] Adapted by Van Liew et al. (2003) and Fernandez et al. (2005).

Table 3. Reported performance ratings for PBIAS.

Model	Value	Performance Rating	Modeling Phase	Reference
HSPF	< 10%	Very good	Calibration and validation	Donigian et al. (1983) ^[a]
HSPF	10% to 15%	Good	Calibration and validation	Donigian et al. (1983) ^[a]
HSPF	15% to 25%	Fair	Calibration and validation	Donigian et al. (1983) ^[a]
SWAT	<15%	Satisfactory	Flow calibration	Santhi et al. (2001)
SWAT	<20%	Satisfactory	For sediment after flow calibration	Santhi et al. (2001)
SWAT	<25%	Satisfactory	For nitrogen after flow and sediment calibration	Santhi et al. (2001)
SWAT	20%	Satisfactory	Calibration and validation	Bracmort et al. (2006)
SWAT	<10%	Very good	Calibration and validation	Van Liew et al. (2007)
SWAT	<10% to <15%	Good	Calibration and validation	Van Liew et al. (2007)
SWAT	<15% to <25%	Satisfactory	Calibration and validation	Van Liew et al. (2007)
SWAT	>25%	Unsatisfactory	Calibration and validation	Van Liew et al. (2007)

^[a] Adapted by Van Liew et al. (2003) and Singh et al. (2004).

Table 4. General performance ratings for recommended statistics for a monthly time step.

Performance Rating	RSR	NSE	PBIAS (%)		
			Streamflow	Sediment	N, P
Very good	$0.00 \leq \text{RSR} \leq 0.50$	$0.75 < \text{NSE} \leq 1.00$	$\text{PBIAS} < \pm 10$	$\text{PBIAS} < \pm 15$	$\text{PBIAS} < \pm 25$
Good	$0.50 < \text{RSR} \leq 0.60$	$0.65 < \text{NSE} \leq 0.75$	$\pm 10 \leq \text{PBIAS} < \pm 15$	$\pm 15 \leq \text{PBIAS} < \pm 30$	$\pm 25 \leq \text{PBIAS} < \pm 40$
Satisfactory	$0.60 < \text{RSR} \leq 0.70$	$0.50 < \text{NSE} \leq 0.65$	$\pm 15 \leq \text{PBIAS} < \pm 25$	$\pm 30 \leq \text{PBIAS} < \pm 55$	$\pm 40 \leq \text{PBIAS} < \pm 70$
Unsatisfactory	$\text{RSR} > 0.70$	$\text{NSE} \leq 0.50$	$\text{PBIAS} \geq \pm 25$	$\text{PBIAS} \geq \pm 55$	$\text{PBIAS} \geq \pm 70$

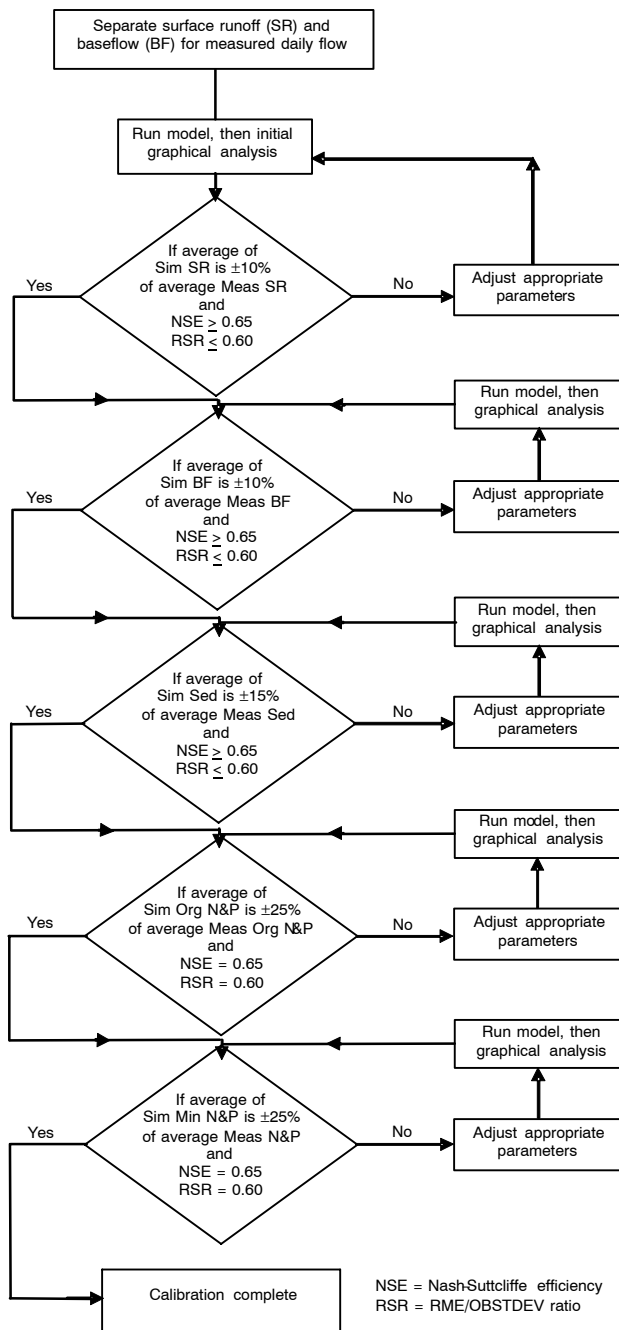


Figure 1. General calibration procedure for flow, sediment, and nutrients in the watershed models (based on calibration chart for SWAT from Santhi et al., 2001).

The next step should be to calculate values for NSE, PBIAS, and RSR. With these values, model performance can be judged based on general performance ratings (table 4). The reported performance ratings and corresponding values developed for individual studies, in addition to the reported

ranges of values (table 1), were used to establish general performance ratings, which appear in table 4. As shown in table 4, the performance ratings for RSR and NSE are the same for all constituents, but PBIAS is constituent specific. This difference is due to the recent availability of information (PBIAS) on the uncertainty of measured streamflow and water quality. Harmel et al. (2006) used the root mean square error propagation method of Topping (1972) to calculate the cumulative probable error resulting from four procedural categories (discharge measurement, sample collection, sample preservation and storage, and laboratory analysis) associated with water quality data collection. Under typical scenarios with reasonable quality control attention, typical financial and personnel resources, and typical hydrologic conditions, cumulative probable error ranges (in similar units to PBIAS) were estimated to be 6% to 19% for streamflow, 7% to 53% for sediment, and 8% to 110% for N and P. These results were used to establish constituent-specific performance rating for PBIAS. Constituent-specific ratings for RSR and NSE can be established when similar information becomes available.

Based on table 4, model performance can be evaluated as “satisfactory” if $\text{NSE} > 0.50$ and $\text{RSR} \leq 0.70$ and, for measured data of typical uncertainty, if $\text{PBIAS} \pm 25\%$ for streamflow, $\text{PBIAS} \pm 55\%$ for sediment, and $\text{PBIAS} \pm 70\%$ for N and P for a monthly time step. These ratings should be adjusted to be more or less strict based on project-specific considerations discussed in the next section.

A general calibration procedure chart (fig. 1) for flow, sediment, and nutrients is included to aid with the manual model calibration process. The recommended values for adequate model calibration are within the “good” and “very good” performance ratings presented in table 4. These limits for adequate manual calibration are stricter than the “satisfactory” rating for general model evaluation because model parameter values are optimized during calibration but not during model validation or application. The importance of and appropriate methods for proper model calibration are discussed in the next section.

ADDITIONAL CONSIDERATIONS

The model evaluation guidelines presented in the previous section apply to the typical case of continuous, long-term simulation for a monthly time step. However, because of the diversity of modeling applications, these guidelines should be adjusted based on single-event simulation, quality and quantity of measured data, model calibration procedure, evaluation time step, and project scope and magnitude.

Single-Event Simulation

When watershed models are applied on a single-event basis, evaluation guidelines should reflect this specific case. Generally, the objectives of single-event modeling are the determination of peak flow rate and timing, flow volume, and recession curve shape (ASCE, 1993; Van Liew et al., 2003).

Accurate prediction of peak flow rate and time-to-peak is essential for flood estimation and forecasting (Ramírez, 2000). Time-to-peak is affected by drainage network density, slope, channel roughness, and soil infiltration characteristics (Ramírez, 2000), and peak flow rate is typically affected by rainfall intensity and antecedent soil moisture content, among other factors.

One of the recommended (ASCE, 1993) model evaluation statistics for the peak flow rate is a simple percent error in peak flow rates (PEP) computed by dividing the difference between the simulated peak flow rate and measured peak flow rate by the measured peak flow rate and expressing the result as a percentage. Model performance related to time-to-peak can be determined similarly. In addition, for single-event simulation, Boyle et al. (2000) recommended that the hydrograph be divided into three phases (rising limb, falling limb, baseflow) based on differing watershed behavior during rainfall and dry periods. A different performance rating should be applied to each hydrograph phase. If the performance ratings are similar for each phase, then a single performance rating can be applied to the overall hydrograph in subsequent evaluation. Otherwise, a different performance rating should be used to evaluate each hydrograph phase.

Quality and Quantity of Measured Data

Although it is commonly accepted that measured data are inherently uncertain, the uncertainty is rarely considered in model evaluation, perhaps because of the lack of relevant data. Harmel et al. (2006) suggested that the uncertainty of measured data, which varies based on measurement conditions, techniques, and constituent type, must be considered to appropriately evaluate watershed models. In terms of the present guidelines and performance ratings, modeled streamflow may be rated “good” if it is within 10% to 15% of measured streamflow data of typical quality (table 4), for example. In contrast, performance ratings should be stricter when high-quality (low uncertainty) data are available. In this case, PBIAS for modeled streamflow may be required to be <10% to receive a “good” rating. If data collected in worst-case conditions are used for model evaluation, then performance ratings should be relaxed to reflect the extreme uncertainty. It can be argued in this case, however, that highly uncertain data offer little value and should not be used in model evaluation.

Finally, in situations when a complete measured time series does not exist, for instance when only a few grab samples per year are available, the data may not be sufficient for analysis using the recommended statistics. In such situations, comparison of frequency distributions and/or percentiles (e.g., 10th, 25th, 50th, 75th, and 90th) may be more appropriate than the quantitative statistics guidelines.

Model Calibration Procedure

Proper model calibration is important in hydrologic modeling studies to reduce uncertainty in model simulations (Engel et al., 2007). Ideal model calibration involves: (1) using data that includes wet, average, and dry years (Gan et al., 1997), (2) using multiple evaluation techniques (Willmott, 1981; ASCE, 1993; Legates and McCabe, 1999; Boyle et al., 2000), and (3) calibrating all constituents to be evaluated.

The calibration procedure typically involves a sensitivity analysis followed by manual or automatic calibration. The most fundamental sensitivity analysis technique utilizes partial differentiation, whereas the simplest method involves

perturbing parameter values one at a time (Hamby, 1994). A detailed review of sensitivity analysis methods is presented by Hamby (1994) and Isukapalli (1999). According to Isukapalli (1999), the choice of a sensitivity method depends on the sensitivity measure employed, the desired accuracy in the estimates of the sensitivity measure, and the computational cost involved. Detailed information on how these factors affect the choice of sensitivity analysis method is given by Isukapalli (1999). After key parameters and their respective required precision have been determined, manual or automatic calibration is done.

Conventionally, calibration is done manually and consists of changing model input parameter values to produce simulated values that are within a certain range of measured data (Balascio et al., 1998). The calibrated model may be used to simulate multiple processes, such as streamflow volumes, peak flows, and/or sediment and nutrient loads. In such cases, two or more model evaluation statistics may be necessary in order to address the different processes (Balascio et al., 1998). However, when the number of parameters used in the manual calibration is large, especially for complex hydrologic models, manual calibration can become labor-intensive (Balascio et al., 1998). In this case automatic calibration is more appropriate.

Automatic calibration involves computation of the prediction error using an equation (objective function) and an automatic optimization procedure (search algorithm) to search for parameter values that optimize the value of the objective function (Gupta et al., 1999). One of the automatic calibration methods is the shuffled complex evolution global optimization algorithm developed at the University of Arizona (SCE-UA) (Duan et al., 1993) and used by vanGriensven and Bauwens (2003) to develop a multi-objective calibration method for semi-distributed water quality models. Other automatic calibration methods include the multilevel calibration (MLC) semi-automated method (Brazil, 1988), the multi-objective complex evolution algorithm (MOCOM-UA) (Yapo et al., 1998), and parameter estimation by sequential testing (PEST) (Taylor and Creelman, 1967), among others.

Data used to calibrate model simulations have a direct effect on the validation and evaluation results. Ideal calibration should use 3 to 5 years of data that includes average, wet, and dry years so that the data encompass a sufficient range of hydrologic events to activate all model constituent processes during calibration (Gan et al., 1997). However, if this is not possible, then the available data should be separated into two sets, i.e., “above-mean” flows (wet years) and “below-mean” flows (dry years), and then evaluated with stricter performance ratings required for wet years (Gupta et al., 1999). Moreover, if the goal is to test the robustness of the model applications under different environmental conditions, then different datasets can be used during model calibration and validation.

In addition, a good calibration procedure uses multiple statistics, each covering a different aspect of the hydrograph, so that the whole hydrograph is covered. This is important because using a single statistic can lead to undue emphasis on matching one aspect of the hydrograph at the expense of other aspects (Boyle et al., 2000). For manual calibration, each statistic should be tracked while adjusting model parameters (Boyle et al., 2000) to allow for balancing the trade-offs in the ability of the model to simulate various aspects of the hydrograph while recognizing potential errors in the observed data.

Finally, ideal model calibration considers water balance (peak flow, baseflow) and sediment and nutrient transport because calibrating one constituent will not ensure that other constituents are adequately simulated during validation. Even though a complete set of hydrologic and water quality data are rarely available, all available data should be considered. To calibrate water balance, it is recommended to separate baseflow and surface flow (surface runoff) from the total streamflow for both the measured and simulated data using a baseflow filter program. The baseflow filter developed by Arnold et al. (1995) and modified by Arnold and Allen (1999) is available at www.brc.tamus.edu/swat/soft_baseflow.html. Baseflow and recharge data from this procedure have shown good correlation with those produced by SWAT (Arnold et al., 2000). With estimated baseflow data, the baseflow ratio can be computed for measured and simulated data by dividing baseflow estimates by the total measured or simulated streamflow. The calibration and validation process can be considered satisfactory if the estimated baseflow ratio for simulated flow is within 20% of the measured flow baseflow ratio (Bracmort et al., 2006). Because plant growth and biomass production can have an effect on the water balance, reasonable local/regional plant growth days and biomass production may need to be verified during model calibration. Annual local/regional evapotranspiration (ET) may also need to be verified or compared with measured estimates during model calibration.

Stricter performance ratings should generally be required during model calibration than during validation. This difference is recommended because parameter values are optimized during model calibration, but parameters are not adjusted in validation, which is possibly conducted under different conditions than those occurring during calibration. Although the importance of model calibration is well established, performance ratings can be relaxed if improper calibration procedures are employed.

It is necessary to note that although proper model calibration is important in reducing error in model output, experience has shown that model simulation results may contain substantial errors. Therefore, rather than provide a point estimate of a given quantity of model output, it may be preferable to provide an interval estimate with an associated probability that the value of the quantity will be contained by the interval (Haan et al., 1998). In other words, uncertainty analysis needs to be included in model evaluations. Uncertainty analysis is defined as the process of quantifying the level of confidence in a given model simulation output based on: (1) the quality and amount of measured data available, (2) the absence of measured data due to the lack of monitoring in certain locations, (3) the lack of knowledge about some physical processes and operational procedures, (4) the approximate nature of the mathematical equations used to simulate processes, and (5) the quality of the model sensitivity analysis and calibration. Detailed model uncertainty analysis is beyond the scope of this research, but more model output uncertainty information can be obtained from published literature.

Evaluation Time Step

Most of the literature reviewed used daily and/or monthly time steps (Saleh et al., 2000; Santhi et al., 2001; Yuan et al., 2001; Sands et al., 2003; Van Liew et al., 2003; Chu and Shirmohammadi, 2004; Saleh and Du, 2004; Singh et al., 2004; Bracmort et al., 2006; Singh et al., 2005; Van Liew et al.,

2007), although a few used annual time steps (Gupta et al., 1999; Shirmohammadi et al., 2001; Reyes et al., 2004), and one used weekly time steps (Narasimhan et al., 2005). Therefore, the time steps considered in this article are the daily and monthly. Typically, model simulations are poorer for shorter time steps than for longer time steps (e.g., daily versus monthly or yearly) (Engel et al., 2007). For example, Yuan et al. (2001) reported an R^2 value of 0.5 for event comparison of predicted and observed sediment yields, and an R^2 value of 0.7 for monthly comparison. The NSE values were 0.395 and 0.656 for daily and monthly, respectively, for DRAINMOD-DUFLOW calibration, and 0.363 and 0.664 for daily and monthly, respectively, for DRAINMOD-W calibration (Fernandez et al., 2005). Similarly, the NSE values were 0.536 and 0.870 for daily and monthly, respectively, for DRAINMOD-DUFLOW validation, and 0.457 and 0.857 for daily and monthly, respectively, for DRAINMOD-W validation (Fernandez et al., 2005). Additional research work that supports the described findings includes that of Santhi et al. (2001), Van Liew et al. (2003), and Van Liew et al. (2007) using SWAT. The performance ratings presented in table 4 for RSR and NSE statistics are for a monthly time step; therefore, they need to be modified appropriately. Generally, as the evaluation time step increases, a stricter performance rating is warranted.

Project Scope and Magnitude

The scope and magnitude of the modeling project also affects model evaluation guidelines. The intended use of the model is an indication of the seriousness of the potential consequences or impacts of decisions made based on model results (U.S. EPA, 2002). For instance, stricter performance rating requirements need to be set for projects that involve potentially large consequences, such as congressional testimony, development of new laws and regulations, or the support of litigation. More modest performance ratings would be acceptable for technology assessment or "proof of principle," where no litigation or regulatory action is expected. Even lower performance ratings will suffice if the model is used for basic exploratory research requiring extremely fast turnaround or high flexibility.

Finally, according to the U.S. EPA (2002), if model simulation does not yield acceptable results based on predefined performance ratings, this may indicate that: (1) conditions in the calibration period were significantly different from those in the validation period, (2) the model was inadequately or improperly calibrated, (3) measured data were inaccurate, (4) more detailed inputs are required, and/or (5) the model is unable to adequately represent the watershed processes of interest.

A CASE STUDY

PROJECT DESCRIPTION

As a component of the CEAP-WAS, the Soil and Water Assessment Tool (SWAT2005; Arnold et al., 1998) was applied to the Leon River watershed in Texas. The watershed drains into Lake Belton, which lies within Bell and Coryell counties and provides flood control, water supply, and public recreation. The lake has a surface area of approximately 12,300 acres with a maximum depth of 124 feet. The total conservation storage is 372,700 ac-ft. The Lake Belton watershed (Leon River) covers approximately 2.3 million acres

within five counties in central Texas. The majority of the land use in the watershed is pasture, hay, and brushy rangeland (63%). Cropland comprises about 10% of the watershed area. The northwestern (upper) half of the watershed contains numerous animal feeding operations, mainly dairies. Currently, there are approximately 60 permitted dairies and 40 smaller dairies (not requiring permits), with approximately 70,000 total cows. The main goal of the study is to use SWAT2005 to predict the impact of land management on the watershed over long periods of time, with special focus on waste management practices.

To accomplish this goal, SWAT2005, with its many input parameters that describe physical, chemical, and biological processes, required calibration for application in the study watershed. Proper model calibration is important in hydrologic modeling studies to reduce uncertainty in model predictions. For a general description of proper model calibration

procedures, refer to the previous discussion in the Additional Considerations section.

MODEL EVALUATION RESULTS BASED ON THE DEVELOPED GUIDELINES

When these model performance ratings were applied to the SWAT2005 modeling in the Leon River (Lake Belton) watershed, the following results were obtained (figs. 2 and 3, table 5). Graphical results during calibration (fig. 2) and validation (fig. 3) indicated adequate calibration and validation over the range of streamflow, although the calibration results showed a better match than the validation results. NSE values for the monthly streamflow calibration and validation ranged from 0.66 to 1.00. According to the model evaluation guidelines, SWAT2005 simulated the streamflow trends well to very well, as shown by the statistical results, which are in agreement with the graphical results. The RSR values ranged

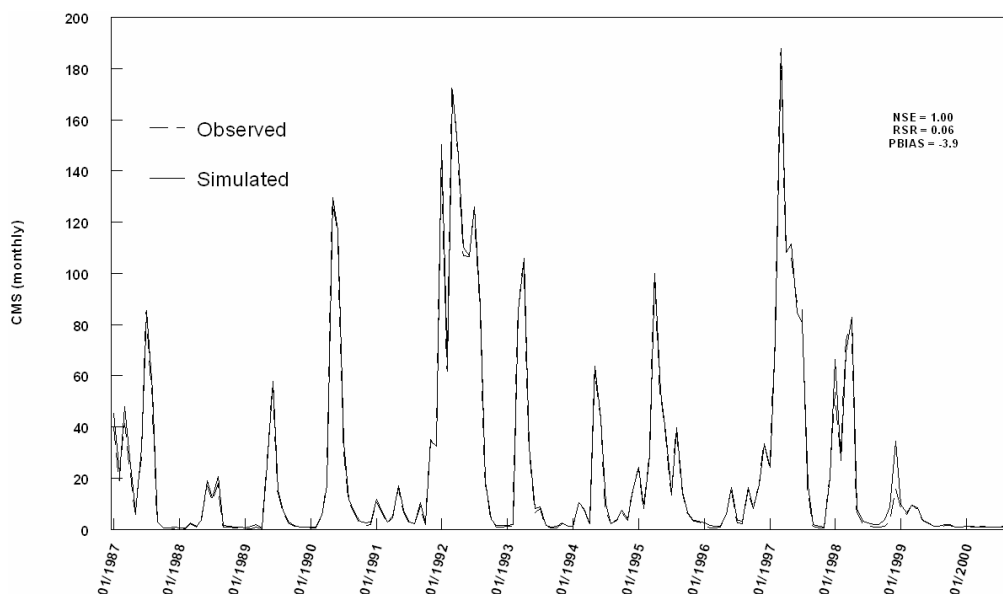


Figure 2. Monthly discharge (CMS) calibration for the Leon River sub-basin 6 WS outlet.

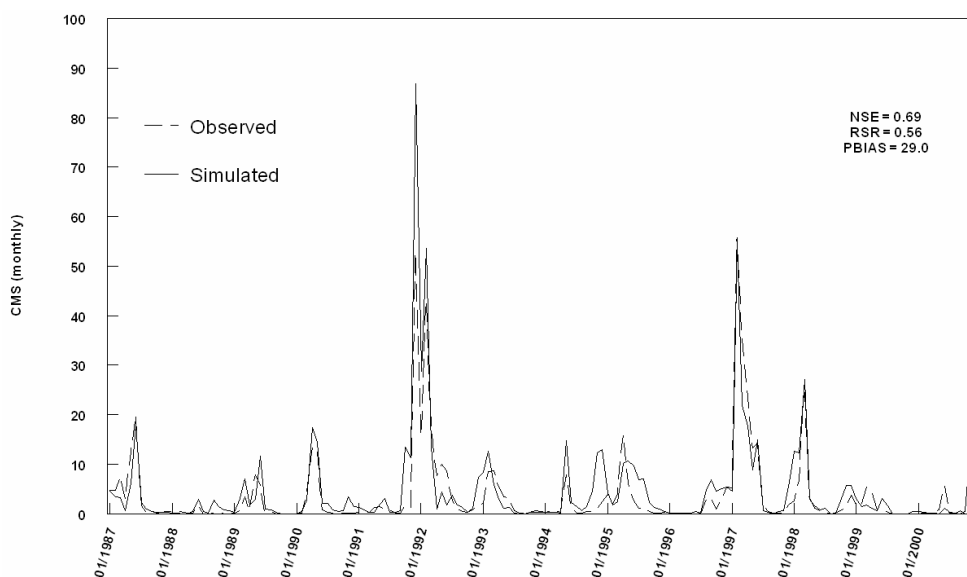


Figure 3. Monthly discharge (CMS) validation for the Leon River sub-basin 13.

Table 5. Results of SWAT2005 average monthly streamflow model output, Leon River watershed, Texas, based on the developed model evaluation guidelines.

Sub-basin	Evaluation Statistic					
	NSE		RSR		PBIAS	
	Calibration	Validation	Calibration	Validation	Calibration	Validation
6	1.00 (very good)	1.00 (very good)	0.03 (very good)	0.06 (very good)	-2.86 (very good)	-3.85(very good)
13	0.66 (good)	0.69 (good)	0.58 (good)	0.56 (good)	-4.89 (very good)	-29.04 (unsatisfactory)
21	0.81 (very good)	0.84 (very good)	0.43 (very good)	0.40 (very good)	-3.62 (very good)	0.41 (very good)
36	0.93 (very good)	0.85 (very good)	0.26 (very good)	0.39 (very good)	-0.28 (very good)	-2.94 (very good)
44	1.00 (very good)	0.78 (very good)	0.06 (very good)	0.46 (very good)	-1.58 (very good)	12.31 (good)
50	0.78 (very good)	--	0.46 (very good)	--	-1.10 (very good)	--
58	0.69 (very good)	--	0.55 (very good)	--	2.15 (very good)	--

from 0.03 to 0.58 during both calibration and validation. These values indicate that the model performance for streamflow residual variation ranged from good to very good. The PBIAS values varied from -4.89% to 2.15% during calibration and from -29.04% to 12.31% during validation. The average magnitude of simulated monthly streamflow values was within the very good range ($PBIAS < \pm 10$) for each sub-basin during calibration (table 5). However, simulated values fell within unsatisfactory, good, and very good ranges during validation for various sub-basins. Aside from one indication of unsatisfactory model performance, SWAT2005 simulation of streamflow was “good” to “very good” in terms of trends (NSE), residual variation (RSR), and average magnitude (PBIAS). As apparent from this evaluation of the Leon River watershed, situations might arise that generate conflicting performance ratings for various watersheds and/or output variables.

In situations with conflicting performance ratings, those differences must be clearly described. For example, if simulation for one output variable in one watershed produces unbalanced performance ratings of “very good” for PBIAS, “good” for NSE, and “satisfactory” for RSR, then the overall performance should be described conservatively as “satisfactory” for that one watershed and that one output variable. However, it would be preferable to describe the performance in simulation of average magnitudes (PBIAS) as “very good,” in simulation of trends (NSE) as “good,” and in simulation of residual variation (RSR) as “satisfactory.” Similarly, if performance ratings differ for various watersheds and/or output types, then those differences must be clearly described.

SUMMARY AND CONCLUSIONS

Most research and application projects involving watershed simulation modeling utilize some type of predefined, project-specific model evaluation techniques to compare simulated output with measured data. Previous research has produced valuable comparative information on selected model evaluation techniques; however, no comprehensive standardization is available that includes recently developed statistics with corresponding performance ratings and applicable guidelines for model evaluation. Thus, the present research selected and recommended model evaluation techniques (graphical and statistical), reviewed published ranges of values and corresponding performance ratings for the recommended statistics, and established guidelines for model evaluation based on the review results and project-specific considerations. These recommendations and discus-

sion apply to evaluation of model simulation related to streamflow, sediments, and nutrients (N and P).

Based on previous published recommendations, a combination of graphical techniques and dimensionless and error index statistics should be used for model evaluation. In addition to hydrographs and percent exceedance probability curves, the quantitative statistics NSE, PBIAS, and RSR were recommended. Performance ratings for the recommended statistics, for a monthly time step, are presented in table 4. In general, model simulation can be judged as “satisfactory” if $NSE > 0.50$ and $RSR \leq 0.70$, and if $PBIAS \pm 25\%$ for streamflow, $PBIAS \pm 55\%$ for sediment, and $PBIAS \pm 70\%$ for N and P for measured data of typical uncertainty. These PBIAS ratings, however, should be adjusted if measurement uncertainty is either very low or very high. As indicated by these PBIAS ratings, it is important to consider measured data uncertainty when using PBIAS to evaluate watershed models. In addition, general guidelines for manual calibration for flow, sediment, and nutrients were presented (fig. 1). Additional considerations, such as single-event simulation, quality and quantity of measured data, model calibration procedure considerations, evaluation time step, and project scope and magnitude, which affect these guidelines, were also discussed. The guidelines presented should be adjusted when appropriate to reflect these considerations. To illustrate the application of the developed model evaluation guidelines, a case study was provided.

Finally, the recommended model evaluation statistics and their respective performance ratings, and the step-by-step description of how they should be used, were presented together to establish a platform for model evaluation. As new and improved methods and information are developed, the recommended guidelines should be updated to reflect these developments.

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APPENDIX

Reported Values of NSE and PBIAS for Various Constituents

Table A-1. Daily and monthly surface runoff calibration and validation value ranges.^[a]

Watershed - Model (Reference)	Statistic	Calibration Value Ranges		Validation Value Ranges	
		Daily	Monthly	Daily	Monthly
Warner Creek, Maryland - SWAT (Chu and Shirmohammadi, 2004)	NSE PBIAS	--	0.35	--	0.77
Black Creek, Indiana - SWAT (Bracmort et al., 2006)	NSE PBIAS	--	0.62 to 0.80	--	0.63 to 0.75

^[a] In tables A-1 through A-9, a dash (--) indicates no value reported for the statistic used; a blank space indicates that the statistic was not used.

Table A-2. Daily and monthly sediment calibration and validation value ranges.

Watershed - Model (Reference)	Statistic	Calibration Value Ranges		Validation Value Ranges	
		Daily	Monthly	Daily	Monthly
Bosque River, Texas - SWAT (Saleh et al., 2000)	NSE PBIAS	--	--	--	0.81
Bosque River, Texas - SWAT (Santhi et al., 2001)	NSE PBIAS	--	0.69 to 0.80	--	0.23 to 0.70
Bosque River, Texas - SWAT (Saleh and Du, 2004)	NSE PBIAS	-2.50	0.83	-3.51	0.59
Bosque River, Texas - HSPF (Saleh and Du, 2004)	NSE PBIAS	0.11	0.72	0.23	0.88
Hellbranch Run, Ohio - HSPF (Engelmann et al., 2002) ^[a]	NSE PBIAS	--	0.49	--	-2.46
Black Creek, Indiana - SWAT (Bracmort et al., 2006)	NSE PBIAS	--	0.86 to 0.92	--	0.68 to 0.75

^[a] In Borah and Bera (2004).

Table A-3. Daily and monthly organic N calibration and validation value ranges.

Watershed - Model (Reference)	Statistic	Calibration Value Ranges		Validation Value Ranges	
		Daily	Monthly	Daily	Monthly
Bosque River, Texas - SWAT (Saleh et al., 2000)	NSE PBIAS	--	--	--	0.78
Bosque River watershed, Texas - SWAT (Santhi et al., 2001)	NSE PBIAS	--	0.57 to 0.58	--	0.43 to 0.73

Table A-4. Daily and monthly NO₃-N calibration and validation value ranges.

Watershed - Model (Reference)	Statistic	Calibration Value Ranges		Validation Value Ranges	
		Daily	Monthly	Daily	Monthly
Bosque River, Texas - SWAT (Saleh et al., 2000)	NSE PBIAS	--	--	--	0.37
Plymouth, North Carolina - DRAINMOD-W (Fernandez et al., 2005)	NSE PBIAS	0.36	0.66	0.46	0.86
Plymouth, North Carolina - DRAINMOD-DUFLOW (Fernandez et al., 2005)	NSE PBIAS	0.40	0.66	0.54	0.87
Bosque River, Texas - SWAT (Santhi et al., 2001)	NSE PBIAS	--	-0.08 to 0.59	--	0.64 to 0.75

Table A-5. Daily and monthly total N (organic N + NO₃-N) calibration and validation value ranges.

Watershed - Model (Reference)	Statistic	Calibration Value Ranges		Validation Value Ranges	
		Daily	Monthly	Daily	Monthly
Bosque River, Texas - SWAT (Saleh et al., 2000)	NSE PBIAS	--	--	--	0.86
Plymouth, North Carolina - DRAINWAT (Amatya et al., 2004)	NSE PBIAS	--	--	0.19	0.76

Table A-6. Daily and monthly PO₄-P calibration and validation value ranges.

Watershed - Model (Reference)	Statistic	Calibration Value Ranges		Validation Value Ranges	
		Daily	Monthly	Daily	Monthly
Bosque River, Texas - SWAT (Saleh et al., 2000)	NSE PBIAS	--	--	--	0.94
Bosque River, Texas - SWAT (Santhi et al., 2001)	NSE PBIAS	--	0.53 to 0.59	--	0.53 to 0.81
Black Creek, Indiana - SWAT (Bracmort et al., 2006)	NSE PBIAS	--	0.78 to 0.84	--	0.51 to 0.74

Table A-7. Daily and monthly organic P calibration and validation value ranges.

Watershed - Model (Reference)	Statistic	Calibration Value Ranges		Validation Value Ranges	
		Daily	Monthly	Daily	Monthly
Bosque River, Texas - SWAT (Saleh et al., 2000)	NSE PBIAS	--	--	--	0.54
Bosque River, Texas - SWAT (Santhi et al., 2001)	NSE PBIAS	--	0.59 to 0.70	--	0.39 to 0.72

Table A-8. Daily and monthly total P calibration and validation value ranges.

Watershed - Model (Reference)	Statistic	Calibration Value Ranges		Validation Value Ranges	
		Daily	Monthly	Daily	Monthly
Black Creek, Indiana - SWAT (Bracmort et al., 2006)	NSE PBIAS	--	0.51	--	0.37

Table A-9. Daily and monthly streamflow calibration and validation value ranges (continued).

Watershed - Model (Reference)	Statistic	Calibration Value Ranges		Validation Value Ranges	
		Daily	Monthly	Daily	Monthly
Bosque River, Texas - SWAT (Saleh et al., 2000)	NSE PBIAS	--	--	--	0.56
Eight watersheds in southwest Oklahoma - SWAT (Van Liew et al., 2003)	NSE PBIAS (continued)	0.56 to 0.58	0.66 to 0.79	-0.37 to 0.72	-1.05 to 0.89

Table A-9 (continued). Daily and monthly streamflow calibration and validation value ranges.

Watershed - Model (Reference)	Statistic	Calibration Value Ranges		Validation Value Ranges	
		Daily	Monthly	Daily	Monthly
Eight watersheds in southwest Oklahoma - HSPF (Van Liew et al., 2003)	NSE PBIAS	0.64 to 0.72	0.74 to 0.82	-1.37 to 0.87	-3.35 to 0.92
Plymouth, North Carolina - DRAINMOD-W (Fernandez et al., 2005)	NSE PBIAS	0.83	0.85	0.87	0.93
Plymouth, North Carolina - DRAINMOD-DUFLOW (Fernandez et al., 2005)	NSE PBIAS	0.68	0.76	0.81	0.92
Bosque River, Texas - SWAT (Santhi et al., 2001)	NSE PBIAS	--	0.79 to 0.83	--	0.62 to 0.87
Six watersheds in Texas - SWAT (Narasimhan et al., 2005) ^[a]	NSE PBIAS	--	0.52 to 0.90	--	0.55 to 0.81
Bosque River, Texas - SWAT (Saleh and Du, 2004)	NSE PBIAS	0.17	0.50	0.62	0.78
Bosque River, Texas - HSPF (Saleh and Du, 2004)	NSE PBIAS	0.72	0.91	0.70	0.86
Plymouth, North Carolina - DRAINWAT (Amatya et al., 2004)	NSE PBIAS	--	--	0.71 to 0.84	0.85
Muscatatuck River, Indiana - SWAT (Vazquez-Amábile and Engel, 2005)	NSE PBIAS	-0.23 to 0.28	0.59 to 0.80	-0.35 to 0.48	0.49 to 0.81
Warner Creek, Maryland - SWAT (Chu and Shirmohammadi, 2004)	NSE PBIAS	--	0.52	--	0.63
Eight CRR-catchment test cases - XNJ (local simplex) (Gan and Biftu, 1996)	NSE PBIAS	0.89 to 0.94 -4.0 to 0.0	-- --	0.37 to 0.88 -16.0 to 34.0	-- --
Eight CRR-catchment test cases - NAM (SCE-UA) (Gan and Biftu, 1996)	NSE PBIAS	0.82 to 0.92 -16.7 to 15.5	-- --	0.41 to 0.88 -31.9 to 37.0	-- --
Eight CRR-catchment test cases - SMAR (SCE-UA) (Gan and Biftu, 1996)	NSE PBIAS	0.86 to 0.91 -20.9 to 12.0	-- --	0.86 to 0.91 -32.9 to 31.7	-- --
Eight CRR-catchment test cases - SMAR (local simplex) (Gan and Biftu, 1996)	NSE PBIAS	0.74 to 0.90 -6.0 to 0.0	-- --	0.00 to 0.85 -27.0 to 44.0	-- --
Eight CRR-catchment test cases - SMAR (local simplex) (Gan and Biftu, 1996)	NSE PBIAS	0.74 to 0.90 -6.0 to 0.0	-- --	0.00 to 0.85 -27.0 to 44.0	-- --
Eight CRR-catchment test cases - XNJ (SCE-UA) (Gan and Biftu, 1996)	NSE PBIAS	0.89 to 0.95 -13.9 to 24.4	-- --	0.45 to 0.88 -31.5 to 30.8	-- --
Eight CRR-catchment test cases - NAM (local simplex) (Gan and Biftu, 1996)	NSE PBIAS	0.81 to 0.92 -4.0 to 0.0	-- --	0.43 to 0.87 -16.0 to 44.0	-- --
Eight CRR-catchment test cases - SMA (SCE-UA) (Gan and Biftu, 1996)	NSE PBIAS	0.87 to 0.94 -7.6 to 1.3	-- --	0.31 to 0.89 -58.6 to 20.1	-- --
Eight CRR-catchment test cases - SMA (local simplex) (Gan and Biftu, 1996)	NSE PBIAS	0.85 to 0.93 -11.9 to 6.4	-- --	0.29 to 0.88 -54.8 to 47.2	-- --
Eight CRR-catchment test cases - SMA (local simplex) (Gan and Biftu, 1996)	NSE PBIAS	0.85 to 0.93 -11.9 to 6.4	-- --	0.29 to 0.88 -54.8 to 47.2	-- --
Eight CRR-catchment test cases - SMA (local simplex) (Gan and Biftu, 1996)	NSE PBIAS	0.85 to 0.93 -11.9 to 6.4	-- --	0.29 to 0.88 -54.8 to 47.2	-- --
Iroquois River, Illinois and Indiana - HSPF (Singh et al., 2004)	NSE PBIAS	0.81	0.88	0.70	0.82
Iroquois River, Illinois and Indiana - SWAT (Singh et al., 2004)	NSE PBIAS	0.79	0.89	0.73	0.83
Ariel Creek, Pennsylvania - SWAT (Peterson and Hamlett, 1998) ^[b]	NSE PBIAS	0.04	0.14	--	--
Ali Efenti, Greece - SWAT (Varanou et al., 2002) ^[b]	NSE PBIAS	0.62	0.81	--	--
University of Kentucky Animal Research Center - SWAT (Spruill et al., 2000) ^[b]	NSE PBIAS	0.19	0.89	-0.04	0.58
Iroquois River, Illinois and Indiana - HSPF (Singh et al., 2005)	NSE PBIAS	0.81	0.88	0.69 to 0.71	0.80 to 0.87

(continued)

Table A-9 (continued). Daily and monthly streamflow calibration and validation value ranges.

Watershed - Model (Reference)	Statistic	Calibration Value Ranges		Validation Value Ranges	
		Daily	Monthly	Daily	Monthly
Iroquois River, Illinois and Indiana - SWAT (Singh et al., 2005)	NSE	0.79	0.88	0.70 to 0.83	0.80 to 0.93
	PBIAS				
Black Creek, Indiana - SWAT (Bracmort et al., 2006)	NSE	--	0.73 to 0.84	--	0.63 to 0.73
	PBIAS				
Five USDA-ARS experimental watersheds - SWAT (Van Liew et al., 2007)	NSE	0.30 to 0.76	0.48 to 0.90	-1.81 to 0.68	-2.50 to 0.89
	PBIAS	2.9 to -91.7	--	2.7 to -155.6	--

[a] Weekly values.

[b] In Borah and Bera (2004).